milliseconds, with no round-trip to a cloud GPU and no dependence on network uptime. This is what makes real-time loss prevention and cold-chain intervention possible at all — a cloud-only inference call would already be too late.

Further upstream, the open lakehouse isn't just cheap storage. Apache Iceberg tables accumulating years of labelled telemetry — shrink events, temperature breaches, shelf-replenishment cycles. These become the training data for the next generation of models. Every open-format, schema-governed event this architecture captures today is a feature set a data science team can use tomorrow, without an ETL rebuild to get it out of a proprietary vendor silo.

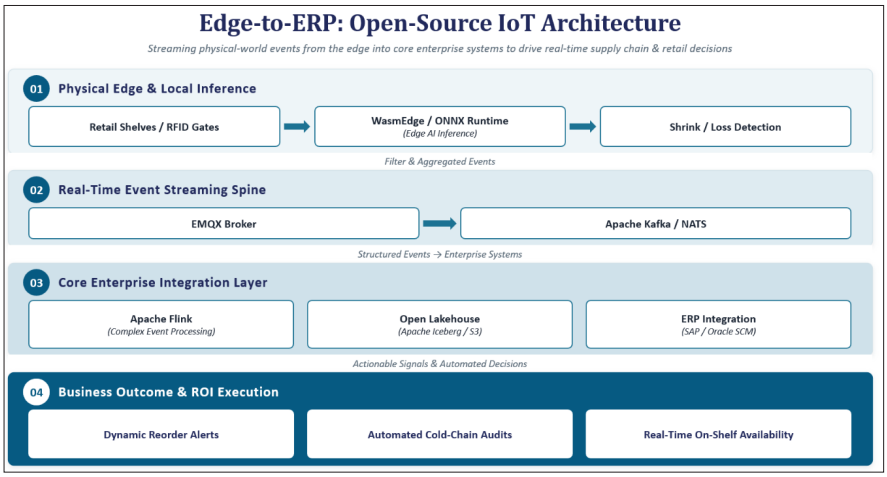


Figure 1: Edge to ERP

The strategic implication for CXOs: an open source IoT stack isn't a parallel investment to an enterprise AI strategy — it's the data and inference substrate that AI strategy depends on. Architectures that lock telemetry into closed platforms don't just pay a cloud tax; they forfeit the training data asset altogether.

High-value industry scenarios: Retail, CPG, and supply chain

Bridging this architectural gap translates directly into bottomline operational efficiency. When edge devices interact seamlessly with core transactional databases, businesses shift from reactive crisis management to proactive automated workflows. Across retail and supply chain ecosystems, two high-impact scenarios demonstrate this value in practice as shown below.

	The business problem	The open source solution	ERP integration and business value
Consumer packaged goods (CPG) and cold chain logistics	Perishable goods (pharmaceuticals, dairy, fresh produce) suffer billions in annual shrinkage due to undetected refrigeration failures during transit. Legacy loggers only record temperature data retroactively upon delivery, making insurance claims difficult and leading to complete inventory write-offs.	Constrained sensor nodes inside refrigerated transport units continuously stream temperature, humidity, and location over lightweight open source messaging brokers (EMQX/Mosquitto). Local micro-inference models — typically lightweight anomaly-detection models (e.g., isolation forests or compact LSTM-based forecasters) small enough to run on constrained edge hardware — running on edge gateways flag temperature drift anomalies in real time.	Apache Flink correlates location streams with SAP supply chain management (SCM) purchase orders. If a cold-chain breach occurs, the system automatically halts the shipment, alerts nearby distribution hubs, and recalculates stock replenishment before store stockouts happen—reducing inventory waste by up to 35%.
Omnichannel retail and dynamic on-shelf availability	Retailers face significant revenue leakage from 'phantom inventory'—situations where the ERP shows an item in stock, but it is misplaced in a backroom or stolen from the sales floor, leading to lost sales and poor customer experiences.	Smart shelves and overhead RFID readers leverage open-source WebAssembly (Wasm) runtimes to process tag reads locally. Edge microservices reconcile shelf counts against local point-of-sale (POS) scans every minute without saturating WAN bandwidth.	Validated stock movement events stream directly into Kafka and update warehouse management systems (WMS). Store associates receive real-time replenishment tasks on mobile devices only when high-margin items fall below critical shelf thresholds, increasing on-shelf availability while optimising labour allocation.